

How dependent is the value of Airbus's A320 Family ramp-up on its suppliers' ability to keep pace?

A scenario-based valuation of the A320 production ramp-up using public
Airbus data

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Independent analysis based on public data. Not affiliated with or endorsed by Airbus.

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Executive Summary

Airbus plans to raise A320 Family production to between 70 and 75 aircraft a month by the end of 2027 and to stabilise at 75 thereafter, against an average of about 51 a month delivered in 2025. The aim is to deliver a backlog of more than 7,000 aircraft that is still growing, so the ramp-up is a necessity more than an expansion. Airbus has postponed the target several times since 2022, and has attributed the delays to its supply chain. This report asks how dependent the value of the ramp-up is on those suppliers' ability to keep pace.

The analysis uses a Python model of yearly deliveries and cash flows built on public Airbus data. Deliveries in each year are limited by Airbus's capacity, by the share of the planned rate that suppliers can support (supply-chain availability) and by demand. The ramp-up is valued against what Airbus's existing factories could deliver without further investment. Airbus does not publish its margin per aircraft, the cost of the ramp-up or its existing capacity, so these are stated assumptions and the results are scenario analysis, not a forecast.

The main findings are as follows.

1. If Airbus follows its target schedule and suppliers support 85% of it, the ramp-up is worth an estimated €1.0 billion in net present value over 2026 to 2035, and €2.5 billion over twenty years. That is worth doing, but with little margin for error. Cumulative cash flow is negative until 2031, with a low point of –€2.7 billion.
2. The 85% is a judgement, because suppliers have not kept pace so far. Against the production rates it planned in 2022 and 2023, Airbus delivered about 77% to 78%, while against its one-year targets it delivered 92% to 102%.
3. The ramp-up pays only if suppliers support at least about 82% of the plan, three points below the base case. Below about 73% the whole investment is lost.
4. Supply-chain availability moves the result more than any other input. Each percentage point is worth about €0.33 billion: the ten-year value is –€0.7 billion at 80%, €2.7 billion at 90% and €4.4 billion at 95%.
5. Delay is nearly as damaging. Each year of unplanned delay costs €0.4 billion to €0.6 billion, and a two-year delay makes the ten-year value negative. A delay that Airbus anticipates, and for which it defers its own investment, costs €0.2 billion to €0.9 billion less.
6. The fastest ramp is worth only €0.1 billion more than one that reaches rate 75 a year later, while a one-year slip costs €0.6 billion. The slower plan has the higher expected value once the chance of such a slip exceeds about 20%.

Three implications follow. Support for suppliers, through financial backing, volume commitments, second sources or bringing work in-house, can be justified on a scale comparable to the investment in Airbus's own factories: five points of availability are worth €1.7 billion, against €2.5 billion assumed for the whole ramp-up. That investment should be released as suppliers demonstrate readiness, not on a fixed calendar. And a slightly slower plan gives up little if the target would have been met and saves considerably more if it would not.

The absolute figures depend on assumptions that cannot be verified from public data and should be treated as indicative. The comparisons between cases are more robust. With demand backed by an order backlog of more than 7,000 aircraft, the return on the capacity Airbus builds is decided mainly by whether its suppliers can feed it.

1. Background

The A320 Family is Airbus's highest-volume programme. It accounted for 607 of the 793 commercial aircraft Airbus delivered in 2025, and Airbus ended that year with a record order backlog of 8,754 commercial aircraft [1]. Airbus is therefore not short of customers. Its challenge is to build aircraft fast enough to serve the orders it already holds.

Its answer is a production ramp-up. In its full-year 2025 results, Airbus said it expects the A320 Family “to reach a rate of between 70 and 75 aircraft a month by the end of 2027, stabilising at rate 75 thereafter” [1], and it repeated that target in its half-year 2026 results [2]. For comparison, 607 deliveries in 2025 is an average of about 51 a month, and the most the programme has delivered in a single year is 642, in 2019 [12] (Figure 1). Rate 75 would mean around 900 aircraft a year.

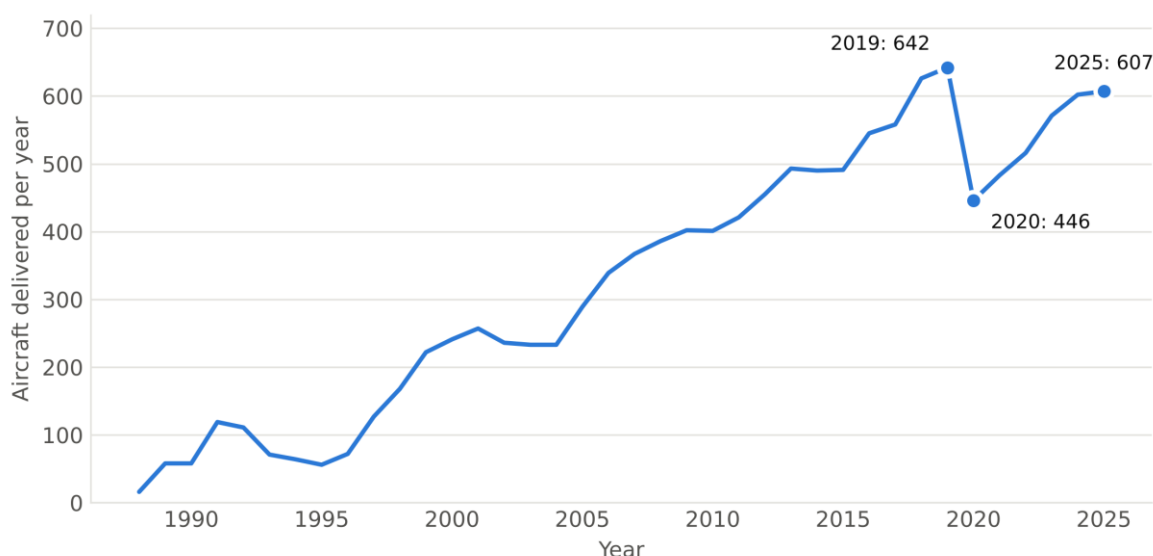


Figure 1. Annual A320 Family deliveries, 1988 to 2025. Deliveries have not yet recovered their 2019 peak. Source: Airbus.

1.1 Why Airbus is ramping up

The ramp-up is better understood as a necessity than as expansion. Airbus's backlog includes more than 7,000 A320neo Family aircraft [8]. At the 607 delivered in 2025, that is more than eleven years of production. The backlog is also still growing: in 2025 Airbus took net orders for 889 commercial aircraft and delivered 793 [1].

Rate 75 would not take Airbus beyond its present position in the market. At 900 aircraft a year it would deliver about 53% of the single-aisle aircraft it expects the market to need [7], in line with the 54% share of single-aisle orders that Airbus already holds with the A320neo Family [11]. The purpose of the ramp-up is to deliver what has already been sold, in a time customers will accept.

Late delivery has a cost. Purchase contracts provide for compensation when an aircraft is delivered late, reported at up to \$20,000 a day for a single-aisle aircraft, although delays caused by suppliers are often treated as excusable and compensation more often takes the form of commercial concessions [13]. A longer queue also makes each new order harder to win. These effects are not included in the model, which counts only the margin on additional aircraft.

1.2 History of the production target

This target has been restated several times. Table 1 sets out what Airbus said about the A320 production rate in successive annual results. The milestone of 65 aircraft a month moved from mid-

2023 to the end of 2024, and rate 75 moved from 2026 to 2027, and then to the current wording of 70 to 75 by the end of 2027.

Table 1. Airbus's stated A320 Family production-rate targets.

Published	Stated target	Source
February 2022	Rate 65 by summer 2023	Financial Statements 2021 [6]
February 2023	Rate 65 by the end of 2024 and rate 75 in 2026	FY2022 results [4]
February 2025	Rate 75 in 2027	FY2024 results [5]
February 2026	Rate 70 to 75 by the end of 2027, stabilising at 75 thereafter	FY2025 results [1]

1.3 Supply-chain constraints

Airbus attributes these revisions to its supply chain, not to demand or to its own factories. In 2023 it said that an adverse operating environment had “prevented our supply chain from recovering at the pace we expected” and that the ramp-up trajectory had “been adapted with suppliers” [4]. In 2026 it was more specific: “Pratt & Whitney’s failure to commit to the number of engines ordered by Airbus is negatively impacting this year’s guidance and the ramp-up trajectory” [1]. Its 2025 financial statements also refer to a “supplier quality issue on fuselage panels impacting its A320 Family programme delivery flow” [3].

The pattern is consistent. An aircraft cannot be delivered without its engines and structures, so the number of aircraft Airbus completes is limited by what its suppliers can provide, whatever the capacity of its own assembly lines. This is the starting point for the analysis: the value of expanding Airbus's capacity depends on whether its suppliers expand with it.

1.4 Information not disclosed by Airbus

Valuing the ramp-up requires three things that Airbus does not publish: the margin it earns on an additional A320 Family aircraft, the investment the ramp-up requires, and the number of aircraft its existing factories could deliver without that investment.

The third matters most for interpreting the results. Airbus has said that deliveries in 2025 were held back by engine supply and a supplier quality issue, so the 607 aircraft delivered that year understate what its existing production system can do. Airbus does not, however, report what that system could deliver if supply were not a constraint. This report therefore assumes a figure, anchored on the 2019 delivery record, and states it as an assumption in the next section. The margin and the investment are treated in the same way.

1.5 Report question and scope

Against this background, the report asks how dependent the value of the A320 Family ramp-up is on suppliers' ability to keep pace. It first values the ramp-up as Airbus has described it, referred to throughout as the Airbus target case. It then tests how much of that value remains if suppliers can support only part of the planned ramp, or if the ramp is delayed, and compares the target with deliberately slower alternatives. The results are scenario analysis based on public data and stated assumptions. They are not a forecast of Airbus's internal economics.

2. Methodology and Assumptions

The analysis uses a Python model of yearly A320 Family deliveries and cash flows from 2026 to 2035. This section summarises how it works. The full assumptions, ramp schedules and formulas are in the Appendix, and the code is available [here](#).

2.1 Model structure

In each year, deliveries are the lowest of three limits (Figure 2). The first is Airbus's own capacity, set by the planned production rate. The second is a supply ceiling: the share of the planned rate that key suppliers can support, referred to as supply-chain availability. At 100%, suppliers scale fully in step with Airbus's plan. At 85%, the base assumption, they support 85% of it. The third is demand, the number of aircraft customers are ready to take.

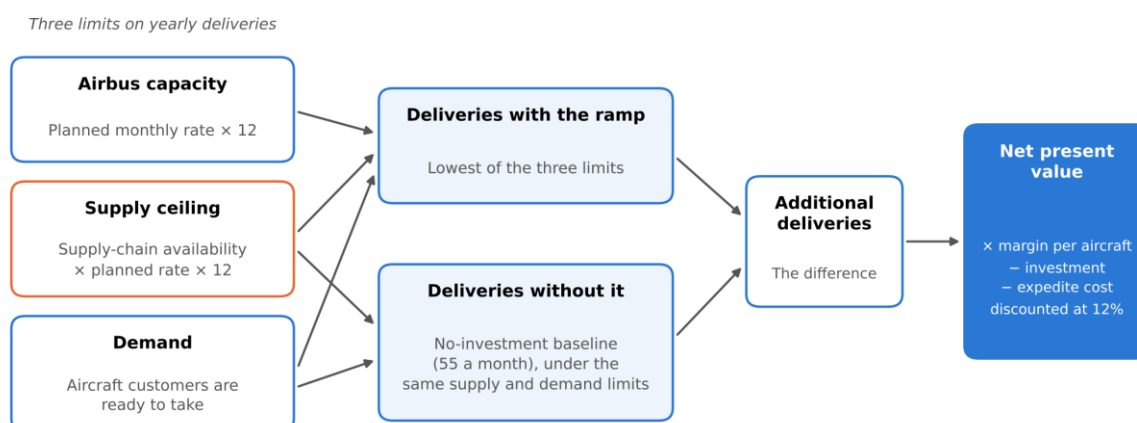


Figure 2. Structure of the model. Source: Own illustration.

The Airbus target case uses a production rate of 58 aircraft a month in 2026, 68 in 2027 and 75 from 2028 onwards, consistent with Airbus's target of 70 to 75 by the end of 2027 and 75 thereafter. Rates are averages over the year, so 2028 is the first full year at rate 75.

2.2 The no-investment baseline

The ramp-up is valued against what Airbus would deliver without it. As Section 1.4 noted, Airbus does not report this figure, so it is an assumption. The model uses 55 aircraft a month. The most the A320 Family has delivered in one year is 642 aircraft in 2019 [12], or 53.5 a month, and Airbus has added capacity since then. The baseline of 55 adds about 3% to that record. Baselines of 53.5 and 60 are tested in Section 6.

The baseline is subject to the same supply and demand limits as the ramp-up. This matters for the report question. If suppliers cannot support more aircraft than the existing factories could already build, the investment adds no deliveries and therefore no value.

2.3 Demand

Delivery demand is assumed to rise to 900 aircraft a year by 2029 and to stay at that level. Three points support this figure. It is the rate at which Airbus itself plans to stabilise production (75 a month). It is about 53% of the single-aisle deliveries Airbus forecasts for the whole market, which is 34,250 aircraft over 2025 to 2044, or roughly 1,700 a year [7], and that share is in line with the A320neo Family's 54% share of single-aisle programme orders [11]. And although it is more than Airbus's net orders across all its commercial programmes in 2024 (826) and 2025 (889) [1], the

difference can be met from a backlog of more than 7,000 A320neo Family aircraft [8]. Demand of 760 and 1,000 aircraft a year is tested in Section 6.

2.4 Valuation

Each additional aircraft delivered, relative to the baseline, earns an assumed margin of €8 million. Against this, the model charges an assumed investment of €2.5 billion to raise capacity from 55 to 75 aircraft a month, which is €125 million for each aircraft a month of added capacity, paid in the year before that capacity is used. Capacity added faster than 5 aircraft a month within one year carries an expedite premium of 50% on that cost, so that a faster ramp is not free. The pace of 5 a month reflects the largest yearly increases the programme has achieved. The premium itself is an assumption, anchored on overtime pay premiums.

Yearly cash flows are discounted at 12%, rounded from the pre-tax weighted average cost of capital of 11.9% that Airbus applies to its commercial aircraft business [3]. The result is the net present value (NPV) of the ramp-up. It is reported in two forms: for 2026 to 2035, and with an extended life in which the 2035 cash flow continues for ten further years. The extension is shown separately because it is an extrapolation. It is more conservative than Airbus's own approach, which adds a terminal value that assumes cash flows grow at 2.6% a year indefinitely [3].

2.5 Key assumptions

Table 2 lists the main inputs. Only the production target, the discount rate and the market forecast come from Airbus. The margin, the investment, the expedite premium, the baseline and supply-chain availability are this report's own assumptions, which is why each is tested over a range in Sections 4 to 6.

Table 2. Key assumptions in the base case and the ranges tested.

Input	Base value	Range tested	Basis
Production rate, Airbus target case	58, 68, then 75 a month from 2028	Delays and slower ramps (Section 5)	Airbus's stated target [1]
No-investment baseline	55 a month	53.5 to 60	Assumption: 2019 delivery record plus about 3%
Supply-chain availability	85%	80% to 90%	Assumption; between the 77% Airbus achieved against earlier plans and 92% to 102% against one-year targets (Section 4.1)
Delivery demand	900 a year	760 to 1,000	Airbus's stabilisation rate; about 53% of its forecast single-aisle market [7]
Margin per additional aircraft	€8 million	€6 to €10 million	Assumption
Ramp-up investment, rate 55 to 75	€2.5 billion	€1.9 to €3.1 billion	Assumption
Expedite premium	50% above 5 a month per year	25% to 100%	Assumption, anchored on overtime pay premiums
Discount rate	12%	10% to 14%	Airbus pre-tax WACC of 11.9% [3]
Valuation period	2026 to 2035, plus 10 further years	Not varied	Assumption; more conservative than Airbus's terminal value [3]

3. Results: Airbus Target Case

If Airbus follows its target schedule and suppliers support 85% of it, the model values the ramp-up at €1.0 billion over 2026 to 2035, and at €2.5 billion with the ten-year extension (Table 3). On these assumptions the ramp-up is worth doing: it earns more than Airbus's cost of capital. There is little margin for error, however. A shortfall of three points in availability, or an unplanned delay of two years, is enough to make the ten-year value negative. This is the case in which Airbus's schedule is met. Sections 4 and 5 test what happens when availability or timing differ.

Table 3. Results for the Airbus target case.

Measure	Result
Deliveries from 2028	765 a year, against 660 without the ramp-up
Additional deliveries, 2026 to 2035	About 870 aircraft
Investment	€2.5 billion, plus €0.4 billion assumed expedite cost
Lowest cumulative cash position	–€2.7 billion, in 2027
Cumulative cash flow positive from	2031
NPV, 2026 to 2035	€1.0 billion
NPV with ten-year extension	€2.5 billion

Two features of the result matter for the rest of the report. The first is that deliveries settle at 765 aircraft a year, not at the 900 that rate 75 implies or that customers would take. The supply ceiling is the binding limit in every year. Airbus pays for capacity of 900 aircraft a year and uses 85% of it, so the investment of 20 rate points yields only 105 additional aircraft a year over the baseline of 660. Each percentage point of availability is worth 9 aircraft a year.

The second is the timing of cash (Figure 3). The investment is made between 2025 and 2027, before the additional aircraft are delivered. In 2026 the ramp-up adds nothing, because suppliers limit deliveries to about 590 aircraft with or without it, and in 2027 it adds only 34. Cumulative cash flow reaches its lowest point of –€2.7 billion in 2027 and does not turn positive until 2031. From 2028 the additional 105 aircraft a year contribute about €0.8 billion a year. With so long a payback, anything that postpones those deliveries, while the investment has already been spent, removes much of the value.

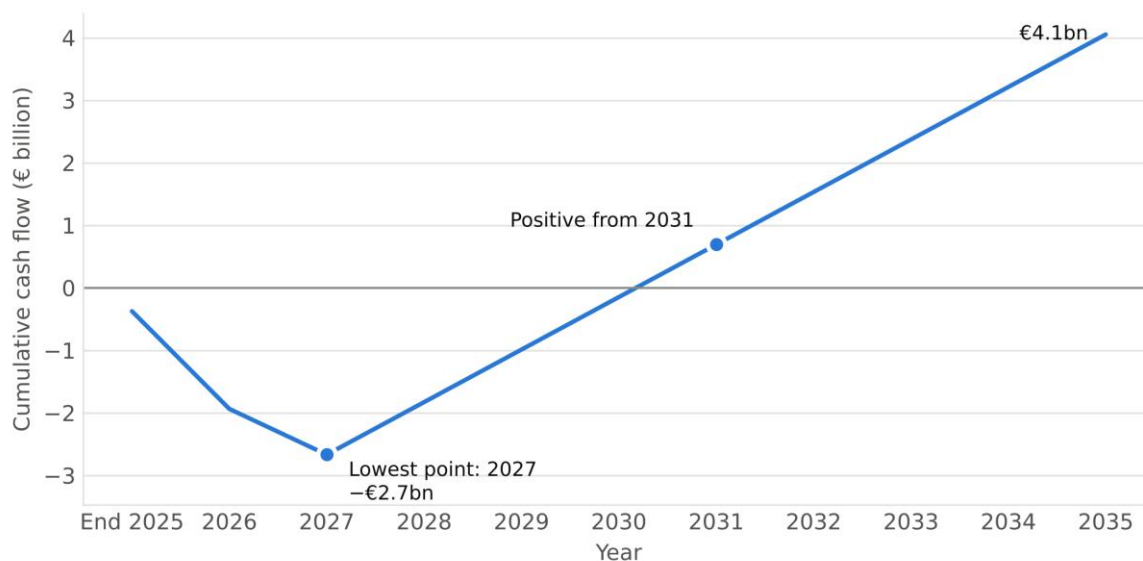


Figure 3. Cumulative cash flow of the ramp-up in the Airbus target case, relative to the no-investment baseline, undiscounted. Cash flow is negative until 2031. Source: Own model.

The target case includes about €0.4 billion of expedite cost, because the rate rises by 10 aircraft a month in 2027 and 7 in 2028, both above the pace of 5 that the model treats as comfortable. This cost is an assumption. Section 5 compares the target with slower ramps that avoid it.

4. Supply-Chain Availability

This section addresses the report question directly. It first sets out what Airbus's suppliers have supported so far, which is the basis for the base assumption of 85%. It then keeps the Airbus target case unchanged and varies only supply-chain availability. Airbus's investment and production plan are the same in every case. Only the suppliers' ability to scale with that plan differs.

4.1 What Airbus has achieved so far

Airbus does not report supply-chain availability, so it has to be inferred. Table 4 compares the A320 Family deliveries implied by Airbus's earlier production targets with what it delivered. In February 2022 Airbus planned to reach rate 65 by summer 2023, which implies about 780 deliveries in 2024. It delivered 602. A year later it planned rate 65 by the end of 2024, which implies at least 780 in 2025. It delivered 607. In both cases Airbus achieved about 77% to 78% of the plan it had set two to three years earlier, and it attributed the shortfall to its supply chain (Section 1.3).

Table 4. A320 Family deliveries against the rate Airbus had planned two to three years earlier.

Plan published	Planned rate	Implied deliveries	Delivered	Share of plan
February 2022 [6]	Rate 65 by summer 2023	About 780 in 2024	602	77%
February 2023 [4]	Rate 65 by the end of 2024	At least 780 in 2025	607	78%

A second measure looks more favourable. Against the delivery target it sets at the start of each year, Airbus achieved 92% to 102% over 2022 to 2025 (Appendix, Table A3). Each of those targets had already been lowered for the supplier constraints Airbus knew about, however, so this measure understates the gap between what Airbus intended and what its suppliers supported.

The base case uses 85%, between the two measures, and Section 6 tests 80% and 90%. Two qualifications apply. The implied deliveries in Table 4 are this report's own calculation, a monthly

rate multiplied by twelve, and not figures published by Airbus. And the shortfall has so far appeared as a postponed target, not as a permanent limit: Airbus still expects to reach rate 75, only later. A lasting availability of 85% and a delay of several years are two ways of representing the same record, and over 2026 to 2035 they give similar values. Section 5 examines delay.

4.2 Value at different levels of availability

The value of the ramp-up rises and falls steadily with availability (Figure 4 and Table 5). At the base assumption of 85% the NPV for 2026 to 2035 is €1.0 billion. At 90% it is €2.7 billion and at 80% it is –€0.7 billion. Each percentage point of availability is worth about €0.33 billion, or about €0.46 billion with the ten-year extension.

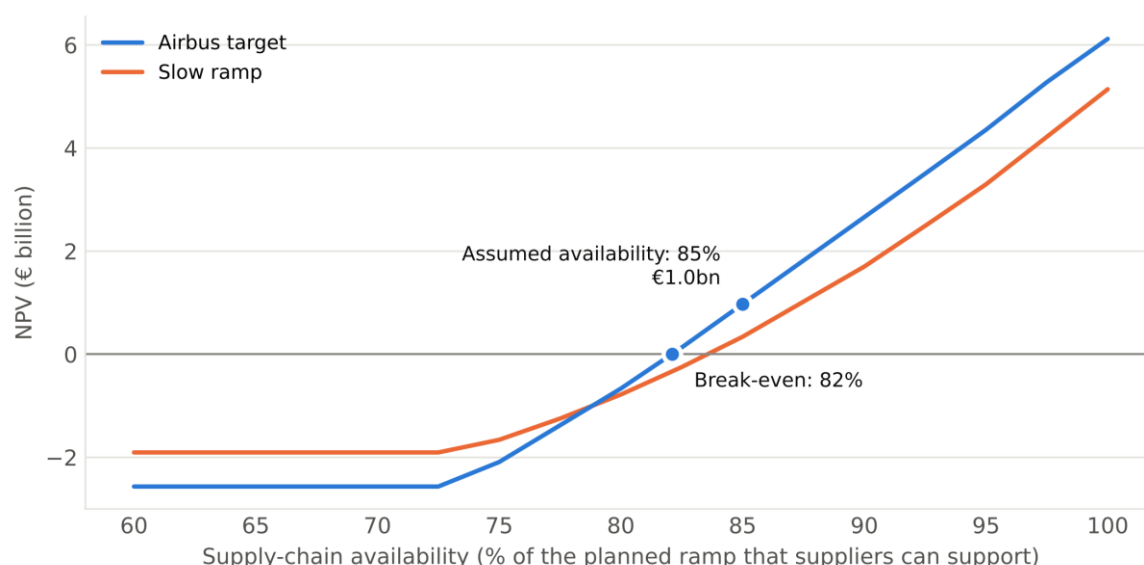


Figure 4. NPV of the ramp-up, 2026 to 2035, by supply-chain availability, for the Airbus target case and the slow ramp. The investment pays only above about 82%. Source: Own model.

Table 5. Airbus target case at different levels of supply-chain availability.

Availability	Deliveries from 2028	NPV, 2026 to 2035	NPV with extension
100%	900 a year	€6.1 billion	€9.6 billion
95%	855 a year	€4.4 billion	€7.2 billion
90%	810 a year	€2.7 billion	€4.8 billion
85% (base)	765 a year	€1.0 billion	€2.5 billion
80%	720 a year	–€0.7 billion	€0.2 billion
78%	702 a year	–€1.2 billion	–€0.6 billion

The ramp-up breaks even at about 82% availability, only three points below the base case. Below that, the additional aircraft do not repay the investment within ten years. At the 77% to 78% that Airbus achieved against its earlier plans, the NPV is about –€1.2 billion to –€1.5 billion, because the ramp-up would add only about 35 to 40 aircraft a year. Below about 73%, suppliers cannot support more aircraft than the existing factories could already build, so the investment adds no deliveries at all and its full cost, about €2.6 billion in present value, is lost.

The upper rows of Table 5 show what is at stake in the other direction. If suppliers supported 95% of the plan, the ramp-up would be worth €4.4 billion over ten years, more than four times the base case.

4.3 Why the value is so sensitive

The investment pays only for the aircraft delivered above the baseline. At 85% availability that is 105 aircraft a year (765 less 660). A shortfall of five percentage points removes 45 aircraft a year, which is 6% of total deliveries but 43% of the additional deliveries, while the investment has already been made in full. A small shortfall in supply therefore becomes a large shortfall in return, and at the base case it is enough to turn the result negative.

A slower ramp does not reduce this exposure. The slow ramp, which reaches rate 75 in 2031, is worth less than the target case at every level of availability above about 79%, and it loses value at a similar rate as availability falls (Figure 4). The supply ceiling applies to the rate Airbus eventually reaches, not only to the years of the ramp, so building up more slowly does not protect the investment if suppliers stop short of the full plan.

4.4 Answer to the report question

The value of the ramp-up is highly dependent on suppliers' ability to keep pace. This holds in two respects. First, whether the ramp-up pays at all depends on suppliers supporting at least about 82% of the plan. Against its earlier plans Airbus has achieved less than that, and the base case clears it by only three points. Second, the amount of value moves sharply with availability: the NPV for 2026 to 2035 is -€0.7 billion at 80%, €1.0 billion at 85% and €2.7 billion at 90%. No other input moves the result as much (Section 6). Airbus can build the capacity, but whether that capacity earns a return, and how much, is decided largely by suppliers it does not control.

Availability is modelled as a single constant percentage. In practice, supply problems arrive as specific disruptions, such as the engine shortfall Airbus described in 2026, and they tend to delay the ramp as well as limit it. Section 5 examines delay separately.

5. Ramp Scenarios

Section 4 asked what happens if suppliers support more or less of the planned rate. This section asks what happens if rate 75 is reached later than Airbus intends, which is what has happened to each earlier target (Table 1). Supply-chain availability stays at 85% throughout. Because a later ramp moves deliveries beyond 2035, the results in this section use the NPV with the ten-year extension, covering 2026 to 2045, unless stated. Over 2026 to 2035 alone, the cost of going later would be overstated.

5.1 Delays

A delay is modelled as the target case shifted back by one to five years, with the rate held at 58 a month in the meantime. Two kinds are distinguished. In a planned delay, Airbus knows in advance and defers the investment along with the ramp. In an unplanned delay, Airbus invests on the target schedule and the rate increases arrive late, so the capacity has been paid for and stands idle. A delay caused by suppliers falling behind is of the second kind.

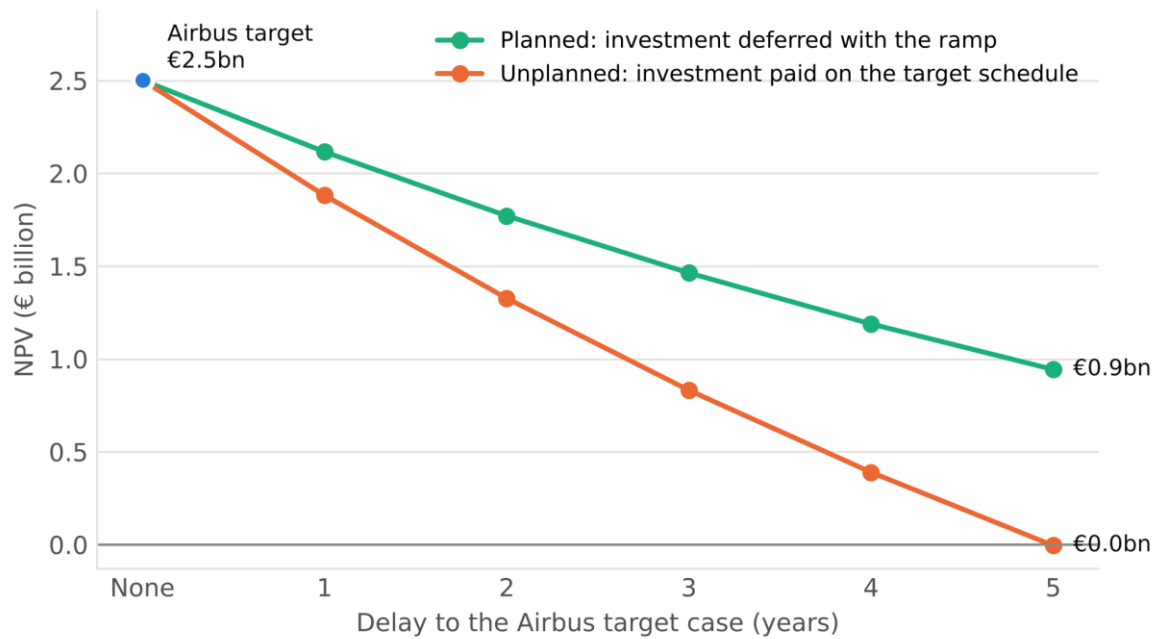


Figure 5. NPV of the ramp-up, 2026 to 2045, by years of delay to the Airbus target case. An unplanned delay of five years removes all of the value. Source: Own model.

An unplanned delay of one year reduces the value from €2.5 billion to €1.9 billion, a quarter of the total, and each further year costs between €0.4 billion and €0.6 billion (Figure 5). After five years, with rate 75 first reached in 2033, nothing remains. Over 2026 to 2035 alone the position is tighter: a one-year unplanned delay leaves €0.4 billion, and a two-year delay makes the NPV negative. One year of unplanned delay costs about as much as a permanent loss of 1.3 percentage points of availability.

The gap between the two lines is the cost of having invested too early. It grows from €0.2 billion for a one-year delay to €0.9 billion for five years. Most of the cost of a delay is the later deliveries themselves, which Airbus cannot avoid, but a large part depends on whether Airbus sees the delay coming in time to hold back its own spending. With a planned delay of five years, €0.9 billion of value remains. With an unplanned one, none does. These figures count only the margin that arrives later. A delay also lengthens the wait for customers who have already ordered (Section 1.1), which the model does not price.

5.2 Slower planned ramps

If delays are likely, Airbus could instead plan a slower ramp from the start. Three alternatives are compared with the target: a moderate, a gradual and a slow ramp, reaching rate 75 in 2029, 2030 and 2031. The moderate ramp raises the rate by 5 or 6 a month in a year, and the other two by no more than 5, so the gradual and slow ramps carry no expedite cost. The yearly rates are listed in the Appendix.

Table 6. Ramp scenarios when each proceeds as planned, at 85% availability.

Scenario	First full year at rate 75	Assumed expedite cost	NPV, 2026 to 2035	NPV, 2026 to 2045
Airbus target	2028	€0.4 billion	€1.0 billion	€2.5 billion
Moderate ramp	2029	€0.1 billion	€0.8 billion	€2.4 billion
Gradual ramp	2030	None	€0.6 billion	€2.1 billion
Slow ramp	2031	None	€0.3 billion	€1.9 billion

When every scenario proceeds as planned, the target case is worth the most (Table 6). Earlier deliveries still outweigh the expedite cost. The margin over the moderate ramp is small, however, at €0.1 billion, and it depends on the assumed expedite premium: at a premium above about 74%, in place of the 50% assumed, the moderate ramp is worth more.

The comparison changes if the target slips. Figure 6 sets each slower ramp beside the target case delayed, unplanned, to the same year. In each pair the slower ramp is worth more, by €0.5 billion to €1.0 billion. A slower ramp still raises the rate every year, pays for capacity only as it is needed and avoids expedite cost. A delayed target does none of these: the rate stays at 58 while the capacity stands idle.

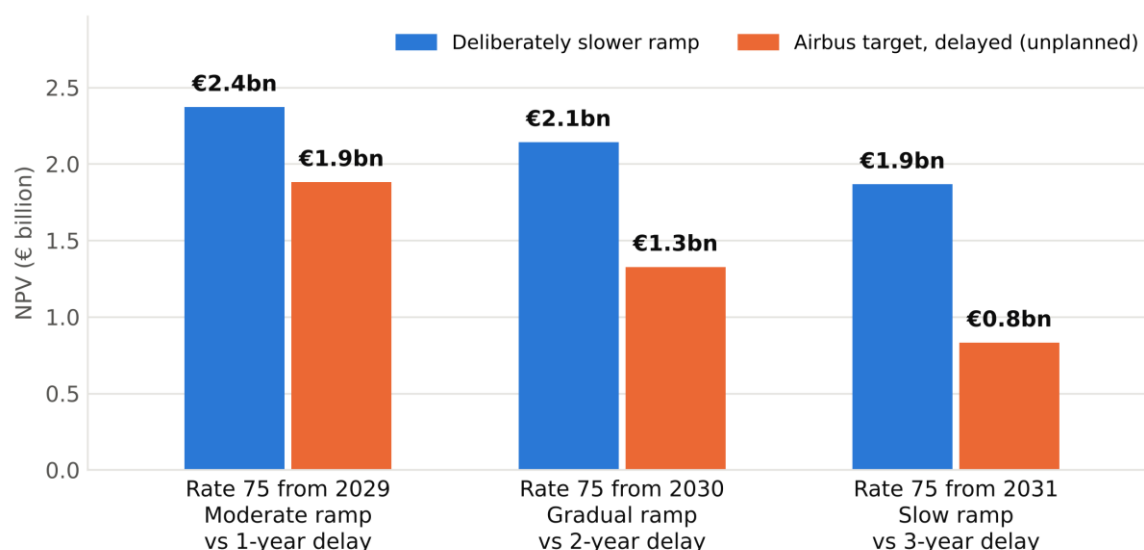


Figure 6. NPV, 2026 to 2045, of deliberately slower ramps compared with the Airbus target case delayed, unplanned, to the same year. Source: Own model.

The choice of ramp is therefore a judgement about how likely the target is to be met. The target is worth €0.1 billion more than the moderate ramp if it is achieved, and €0.5 billion less if it slips by a year. On these figures, the moderate ramp has the higher expected value once the chance of a one-year slip exceeds about 20%, assuming the moderate ramp itself is achieved. Given how often the target has moved since 2022, that threshold is not a demanding one.

6. Sensitivity Analysis

This section tests all the main assumptions together, to show how far the conclusions depend on values that Airbus does not publish. Each input is moved to its downside and its upside value in turn, with everything else held at the base case (Figure 7). Results are the NPV of the Airbus target case for 2026 to 2035, which is €1.0 billion in the base case.

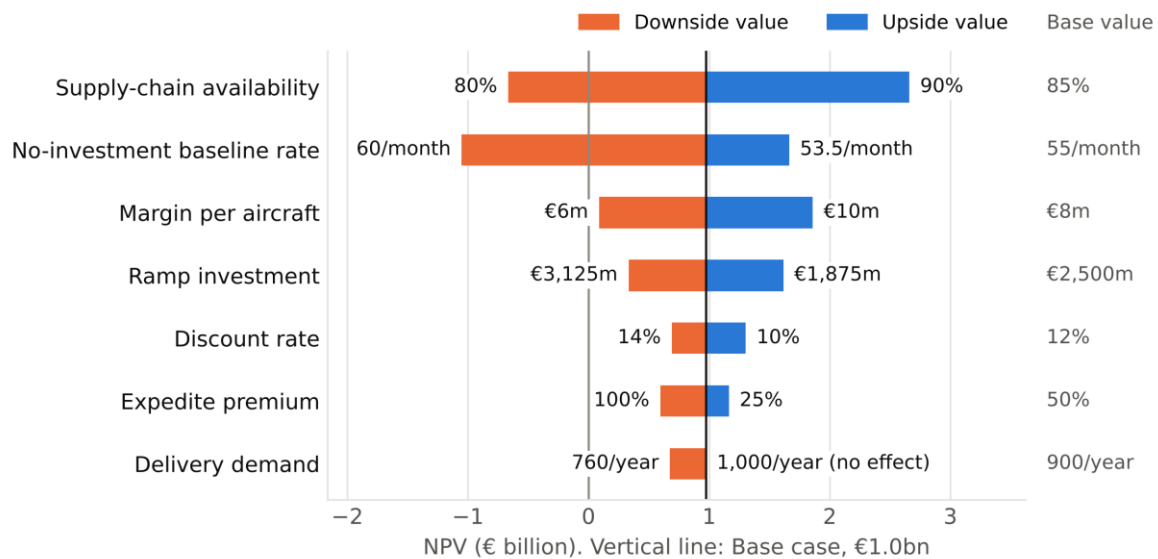


Figure 7. NPV of the Airbus target case, 2026 to 2035, when one assumption at a time is moved to its downside or upside value. Supply-chain availability has the largest effect. Source: Own model.

6.1 Inputs that matter most

Supply-chain availability has the largest effect. Moving it by five points in either direction takes the NPV from –€0.7 billion to €2.7 billion, a range of €3.3 billion. The range tested, five points either side of the base, is narrower than the gap between the two measures of Airbus's record in Section 4.1.

The no-investment baseline comes second, with a range from –€1.1 billion to €1.7 billion. If the existing factories could already deliver 60 a month, the ramp-up would add only 45 aircraft a year at 85% availability and would lose money. This input differs from the others in that it is not uncertain to Airbus, only to an outside observer. It is the main reason the absolute NPV figures in this report should be treated as indicative.

The margin per aircraft moves the NPV between €0.1 billion and €1.9 billion, and the ramp-up investment between €0.3 billion and €1.6 billion. Because the base value is small, neither can be ignored: a 25% overrun on the investment removes about two thirds of it.

6.2 Inputs that matter less

The discount rate and the expedite premium each move the NPV by €0.4 billion or less in either direction. Delivery demand also matters little. At 85% availability Airbus can deliver only 765 aircraft a year, so demand of 760 a year instead of 900 costs €0.3 billion, and demand above 900 adds nothing. While suppliers hold deliveries below what customers would take, the strength of demand is not what limits the value of the ramp-up.

6.3 Combined cases

Moving one input at a time understates the range of outcomes, because adverse conditions tend to occur together. With every input at its downside value, the NPV for 2026 to 2035 is –€3.5 billion: the ramp-up adds no deliveries and the whole investment is lost. With every input at its upside value, it is €6.3 billion. Neither is likely, but the spread shows how wide the outcomes are around a base case that is only slightly positive.

6.4 Implication for the report question

The two inputs with the largest effect, availability and the baseline, both determine how many additional aircraft the ramp-up delivers. Of the two, the baseline is known to Airbus even though it is

not published, which leaves availability as the largest source of real uncertainty in the result. The cost assumptions and the discount rate matter less, and demand hardly at all. This is consistent with the finding of Section 4: the value of the ramp-up is highly dependent on what suppliers can deliver.

7. Strategic Implications

The thin margin in the base case is not an argument against the ramp-up. Airbus has little choice: the aircraft are already sold, and the backlog cannot be delivered in an acceptable time at today's rate (Section 1.1). The practical question is therefore not whether to ramp up but how to make the ramp-up pay, and the results point to four implications. They follow from the model's assumptions and should be read with the limitations in Section 8.

7.1 Supplier capability is worth paying for

Each percentage point of supply-chain availability is worth about €0.33 billion over 2026 to 2035. Raising availability from 85% to 90% adds €1.7 billion, which is two thirds of the €2.5 billion assumed for the whole of Airbus's own capacity investment, and it nearly triples the value of the ramp-up. Spending that helps suppliers keep pace, such as financial support, longer-term volume commitments, second sources for critical parts or bringing work in-house, can therefore be justified on a scale comparable to the investment in Airbus's own factories. Airbus has already acted in this direction: in December 2025 it completed the acquisition of Spirit AeroSystems businesses that supply it, including wing component production for the A320 [3]. The model supports the logic of such moves, although it does not value any of them individually.

7.2 Investment should be paced by supplier readiness

Capacity that arrives before suppliers can feed it earns nothing. A delay that Airbus anticipates, and for which it holds back its own spending, costs €0.2 billion to €0.9 billion less than the same delay discovered after the capacity has been paid for (Section 5.1). With a base value of €2.5 billion over twenty years, that difference is large. It favours releasing investment in stages that are tied to firm supplier commitments, engine deliveries above all, over committing to the full programme on a fixed calendar.

7.3 A slightly slower plan may be worth more

The target case is the most valuable ramp if it is achieved, but only by €0.1 billion over the moderate ramp, while a one-year slip costs €0.6 billion. The pay-off is asymmetric: there is little to gain from the faster plan and much more to lose if it is missed. On the figures in Section 5.2, planning for rate 75 one year later has the higher expected value once the chance of a one-year slip exceeds about 20%. Airbus's current wording, a rate of 70 to 75 by the end of 2027, already leaves some of this room.

7.4 More capacity is not the next constraint

In the base case, deliveries stop at 765 aircraft a year against demand of 900 and installed capacity of 900. The shortfall of 135 aircraft a year is caused by supply, not by Airbus's factories or its order book. Once rate 75 is in place, the next gain comes from raising availability, not from adding further capacity.

8. Limitations

The analysis is built on public data and a deliberately simple model. Five limitations affect how the results should be read.

Key inputs are assumptions. The margin per aircraft, the ramp-up investment, the expedite premium and the no-investment baseline are not published by Airbus. The absolute NPV figures are therefore indicative, and because the base value is small, modest changes in these inputs can change its sign. The comparisons between cases, such as the value of a point of availability or the cost of a year's delay, are more robust, because the same assumptions apply on both sides.

Supply-chain availability is a single number. The base value of 85% is a judgement between two measures of Airbus's record, not a measured quantity. It is constant over time and treats all suppliers as one. In practice the constraint is a small number of specific parts, engines above all, and it varies from year to year. The model also assumes that suppliers support a fixed share of whatever rate Airbus plans, which is a simplification of how supplier capacity is actually built.

The economics of each aircraft are constant. Every additional aircraft earns the same margin. The model does not capture learning effects, changes in the mix towards the A321, price escalation, compensation paid to customers for late deliveries, or tax. It also assumes Airbus builds only what it can deliver, so it does not capture the cost of undelivered aircraft waiting for engines or the working capital tied up in the ramp-up.

Scenarios are stylised. Rates are yearly averages. A delay holds the rate at 58 a month and then follows the target path, whereas a real delay would be more gradual. The slower ramps are assumed to be achieved as planned, which favours them in the comparison with a delayed target. The model's deliveries for 2026, about 590 aircraft or 49 a month, are also above Airbus's pace in the first half of that year, when it delivered 271 A320 Family aircraft, about 45 a month [2]. This has no effect on the results, because 2026 adds no deliveries above the baseline.

The scope is Airbus's own cash flows on one programme. The investment that suppliers must make to keep pace, and who pays for it, is outside the model. So are the wider costs of delivering late: compensation and concessions to customers, orders that are harder to win while the queue is long, and credibility with customers and investors. Because the model counts only the margin on additional aircraft, it probably understates the case for ramping up, and with it the cost of suppliers falling short. Demand follows a fixed path with no cancellations.

9. Conclusion

This report asked how dependent the value of Airbus's A320 Family ramp-up is on its suppliers' ability to keep pace. The answer is that it is highly dependent on them, both for whether the ramp-up pays at all and for how much it is worth. This matters all the more because the ramp-up is a necessity: Airbus needs it to deliver a backlog of more than 7,000 aircraft, and cannot simply decline to invest if the return looks thin.

Against the production rates it planned in 2022 and 2023, Airbus delivered about 77% to 78%. If suppliers support 85% of the current plan, the ramp-up is worth an estimated €1.0 billion over 2026 to 2035 and €2.5 billion over twenty years. That is worth doing, but with little margin for error. It pays only above about 82% availability, three points below that assumption, and below about 73% the whole investment is lost.

No other input moves the result as much. Each percentage point of availability is worth about €0.33 billion, so the range from 80% to 90% spans –€0.7 billion to €2.7 billion. Each year of unplanned delay costs €0.4 billion to €0.6 billion, and five years of it removes the value entirely. Demand, by contrast, hardly matters while suppliers hold deliveries below what customers would take.

Three implications follow. Support for suppliers can be justified on a scale comparable to the investment in Airbus's own factories. That investment should be released as suppliers demonstrate

readiness, not on a fixed calendar. And a plan that reaches rate 75 a year later gives up little if the target would have been met and saves considerably more if it would not.

These figures rest on assumptions about margins, investment and existing capacity that Airbus does not publish, and they should be read as scenario analysis, not as a forecast. The direction of the finding is less sensitive to those values than the figures are: with demand secured by a backlog of this size, the value of adding capacity is decided by the slowest part of the supply chain.

References

- [1] Airbus (2026). Airbus reports Full-Year (FY) 2025 results. Press release, 19 February 2026.
- [2] Airbus (2026). Airbus reports Half-Year (H1) 2026 results. Press release, 29 July 2026.
- [3] Airbus (2026). Financial Statements 2025. Note 8, Acquisitions and Disposals; Note 19, Intangible Assets; Note 23, Inventories.
- [4] Airbus (2023). Airbus reports Full-Year (FY) 2022 results. Press release, 16 February 2023.
- [5] Airbus (2025). Airbus reports Full-Year (FY) 2024 results. Press release, 20 February 2025.
- [6] Airbus (2022). Financial Statements 2021.
- [7] Airbus (2025). Global Market Forecast 2025–2044. Press release, 12 June 2025.
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- [10] Airbus (2024). Airbus reports Full-Year (FY) 2023 results. Press release, 15 February 2024.
- [11] Leeham News and Analysis (2026). Airbus dominates single-aisle sales as Boeing wins with widebodies. Aviation International News, 27 January 2026.
- [12] Airbus (2020). Airbus delivers strong 2019 commercial aircraft performance. Press release, 10 January 2020.
- [13] Reuters (2023). Airbus and Boeing jet delivery delays spark penalties battle. Published by CNBC, 18 January 2023.

Appendix

This appendix gives the full inputs and calculations behind the results. The code and data files are available [here](#).

A. Reported Airbus data used

Table A1. Figures taken from Airbus publications.

Item	Value	Source
A320 Family deliveries, 2025	607	[1]
Commercial aircraft deliveries, 2025	793	[1]
Commercial aircraft order backlog, end of 2025	8,754	[1]
Highest yearly A320 Family deliveries	642 (2019)	[12]
A320 Family production target	70 to 75 a month by the end of 2027, stabilising at 75	[1], [2]
Pre-tax weighted average cost of capital, commercial aircraft	11.9%	[3], Note 19
Perpetual growth rate used by Airbus for terminal value	2.6%	[3], Note 19
Single-aisle deliveries forecast, 2025 to 2044	34,250	[7]
A320neo Family share of single-aisle programme orders	54%	[11]

B. Model assumptions

Table A2 lists every assumption with its downside and upside value. None of these values is Airbus data. The downside and upside values are used in the sensitivity analysis in Section 6.

Table A2. Model assumptions.

Assumption	Downside	Base	Upside	Basis
Margin per additional aircraft	€6m	€8m	€10m	Illustrative; range of plus or minus 25%
Ramp-up investment, rate 55 to 75	€3,125m	€2,500m	€1,875m	Illustrative; range of plus or minus 25%
No-investment baseline rate	60 a month	55 a month	53.5 a month	Upside is the 2019 record of 642 deliveries; base adds about 3% for capacity added since
Supply-chain availability	80%	85%	90%	Judgement between two measures: Airbus delivered about 77% of the rate planned two to three years earlier (Table 4) and 92% to 102% of its one-year targets in 2022 to 2025 (Table A3). Range of plus or minus five points
Long-run delivery demand	760 a year	900 a year	1,000 a year	See Section C
Comfortable yearly rate increase	Not varied	5 a month per year	Not varied	Largest yearly rise in A320 Family deliveries since 2000 was 5.7 a month (2018)
Expedite premium	100%	50%	25%	Anchored on overtime pay premiums of 25% and 50% under French labour law; 100% allows for lost productivity under sustained overtime
Discount rate	14%	12%	10%	Rounded from Airbus's pre-tax WACC of 11.9% [3]; range of plus or minus two points
Extended life after 2035	Not varied	10 years	Not varied	Final-year cash flow continued with no growth

Table A3 shows the more favourable of the two measures behind the availability assumption: Airbus's commercial aircraft deliveries against the target set at the start of each year. The targets cover all commercial aircraft, not only the A320 Family, and each already reflects the supplier constraints known when it was set (Section 4.1).

Table A3. Airbus commercial aircraft deliveries against start-of-year targets.

Year	Target	Delivered	Delivered as share of target	Source
2022	720	661	92%	[9], [4]
2023	720	735	102%	[4], [10]
2024	Around 800	766	96%	[10], [5]
2025	Around 820	793	97%	[5], [1]

C. Demand

Delivery demand is the number of aircraft customers are assumed ready to take in a year. It rises to 900 a year by 2029 (Table A4). Deliveries are also limited by the order backlog, which starts at the end-2025 level, falls with deliveries and rises with new orders. The backlog does not become a binding limit in any scenario. In the downside and upside demand cases, delivery demand in every year is scaled in proportion, to 760 and 1,000 a year in the long run.

Table A4. Base demand plan, aircraft a year.

	2026	2027	2028	2029 to 2035
Delivery demand	800	840	870	900
New orders	800	820	820	820

The long-run level of 900 a year rests on three points. First, it equals rate 75, at which Airbus plans to stabilise production [1]. Second, Airbus forecasts 34,250 single-aisle deliveries across the market over 2025 to 2044, about 1,700 a year [7], so 900 a year is a share of about 53%, in line with the A320neo Family's 54% share of total programme orders in the single-aisle sector at the end of 2025 [11]. Third, although 900 a year is above Airbus's net orders across all its commercial programmes in 2024 (826) and 2025 (889) [1], the difference can be met for many years from a backlog of more than 7,000 A320neo Family aircraft [8]. The downside value of 760 a year is closer to recent order intake, so that deliveries no longer rely on drawing down the backlog.

D. Ramp scenarios

Table A5. Assumed production rate by scenario, aircraft a month, yearly average.

Scenario	2026	2027	2028	2029	2030	2031 to 2035
Airbus target	58	68	75	75	75	75
Moderate ramp	58	64	70	75	75	75
Gradual ramp	58	62	66	71	75	75
Slow ramp	58	61	65	68	72	75

In a delay case, the rate stays at 58 a month for the number of years of the delay and then follows the Airbus target path. In a planned delay the investment moves with the ramp. In an unplanned delay the investment follows the target schedule (Table A6).

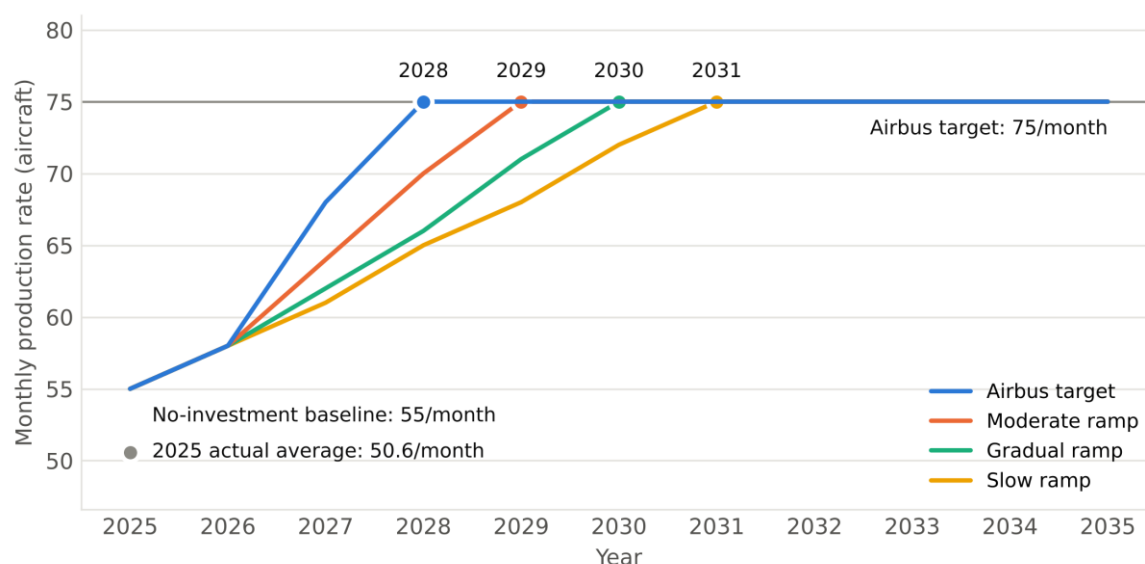


Figure A1. Assumed production rate by ramp scenario. Source: Own model.

Table A6. NPV of the Airbus target case by years of delay, 2026 to 2045.

Delay	First full year at rate 75	Planned delay	Unplanned delay
None	2028	€2.5 billion	€2.5 billion
1 year	2029	€2.1 billion	€1.9 billion
2 years	2030	€1.8 billion	€1.3 billion
3 years	2031	€1.5 billion	€0.8 billion
4 years	2032	€1.2 billion	€0.4 billion
5 years	2033	€0.9 billion	€0.0 billion

E. Calculations

For each year from 2026 to 2035, the model calculates:

$\text{supply ceiling} = \text{planned monthly rate} \times 12 \times \text{supply-chain availability}$
 $\text{deliveries} = \text{lowest of } (\text{planned rate} \times 12, \text{supply ceiling, delivery demand, backlog})$
 $\text{baseline deliveries} = \text{lowest of } (\text{baseline rate} \times 12, \text{supply ceiling, delivery demand})$
 $\text{additional deliveries} = \text{deliveries} - \text{baseline deliveries}$
 $\text{cash flow} = \text{additional deliveries} \times \text{margin per aircraft} - \text{investment} - \text{expedite cost}$

Investment is €125 million for each aircraft a month of capacity above the baseline (€2,500 million divided by the 20 rate points between 55 and 75). It is paid in the year before the capacity is used, so the first payment falls at the end of 2025. Where the rate rises by more than 5 a month within one year, the rate points above 5 cost an additional 50%. In the Airbus target case this applies to 5 rate points in 2027 and 2 in 2028, which gives an expedite cost of €438 million.

The value measures are:

$\text{NPV, 2026 to 2035} = - \text{initial investment} + \text{sum of cash flow in year } t \div (1 + \text{discount rate})^t, \text{ for } t = 1 \text{ to } 10$
 $\text{extension value} = \text{sum of 2035 cash flow} \div (1 + \text{discount rate})^t, \text{ for } t = 11 \text{ to } 20$
 $\text{NPV, 2026 to 2045} = \text{NPV, 2026 to 2035} + \text{extension value}$

The model caps production at what can be delivered, so no undelivered aircraft accumulate. Cash flows are before tax, in constant euros.

F. Additional figures

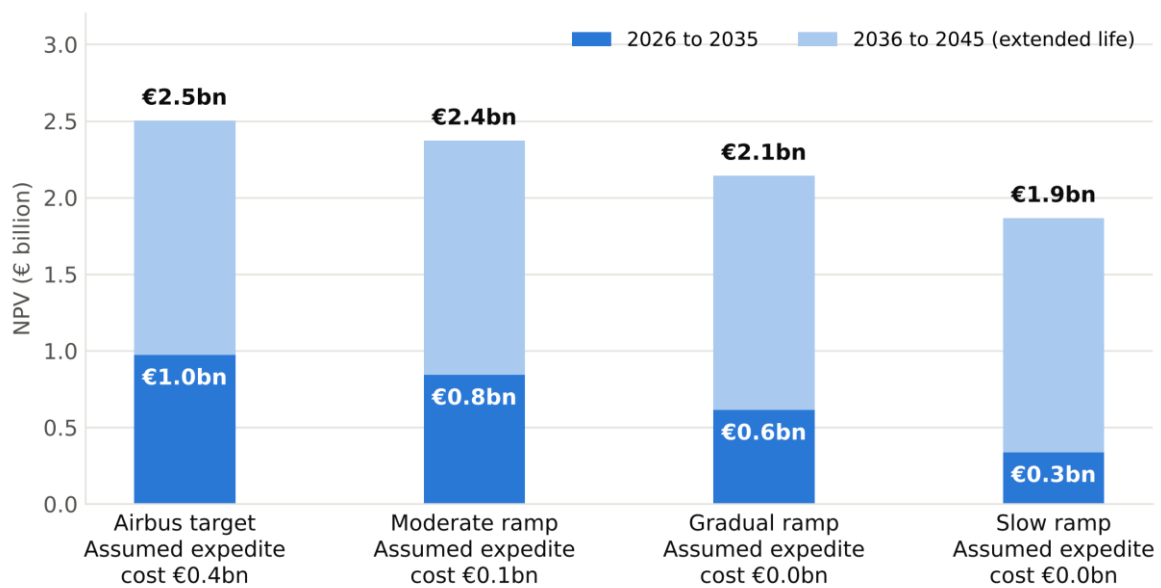


Figure A2. NPV by ramp scenario at 85% availability, showing the 2026 to 2035 value and the ten-year extension. Source: Own model.

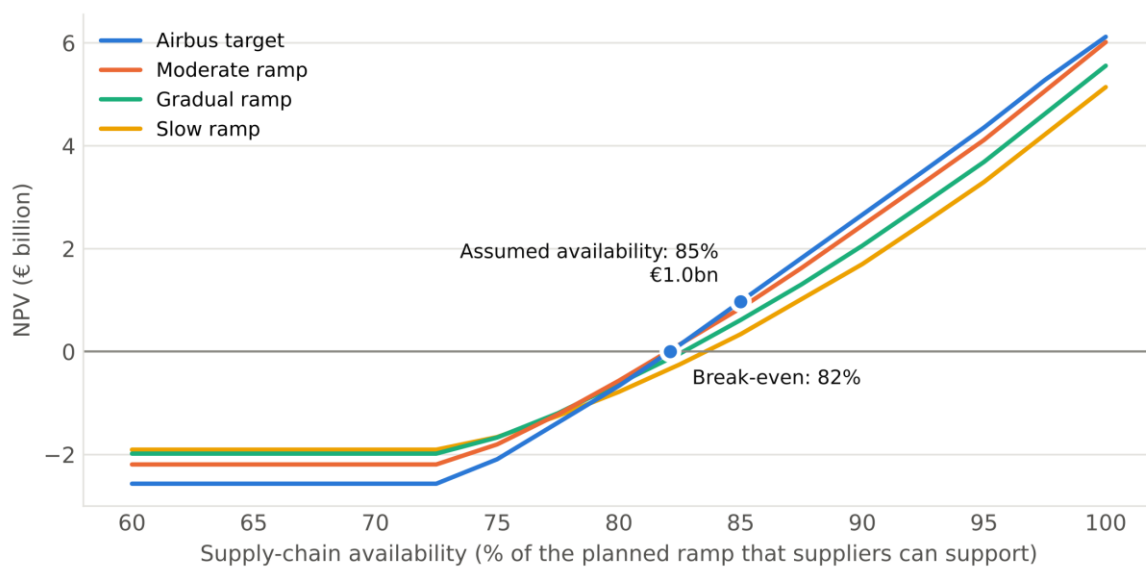


Figure A3. NPV, 2026 to 2035, by supply-chain availability for all four ramp scenarios. Source: Own model.

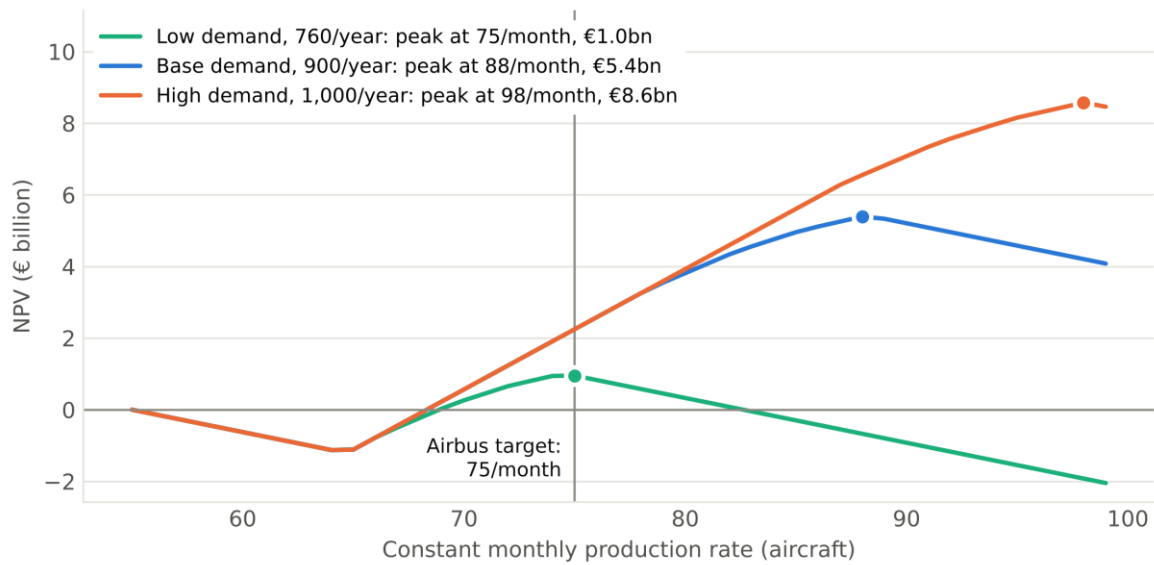


Figure A4. NPV, 2026 to 2035, by production rate for three levels of demand. This figure uses a simplified case in which the rate is constant from the first year and all investment is paid at the start, so its values are higher than those of the ramp scenarios. The value peaks above rate 75 because the model assumes suppliers support 85% of whatever rate is planned, so a higher planned rate raises the supply ceiling. If supplier output were fixed, planning a higher rate would add cost and no deliveries. Source: Own model.

G. GitHub

<https://github.com/IshaanFartya/Airbus-A320-Capacity-Analysis/tree/v1.0>